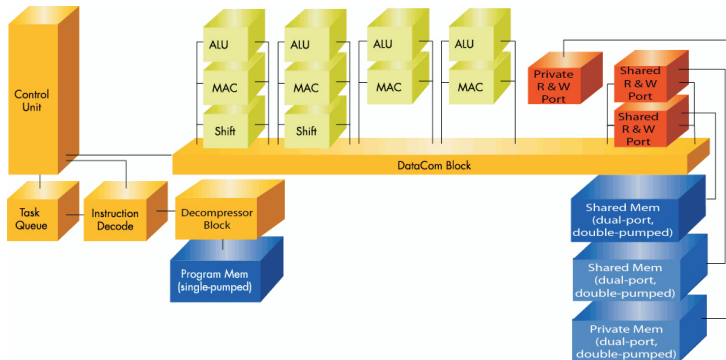


JAZZ DSP CORES

JAZZ DSP PROCESSORS



The Jazz DSP cores are highly optimized DSPs designed for general-purpose, fixed-point DSP programming. Whether you are replacing an existing DSP core or adding new DSP capabilities to your design, the Jazz DSP cores can get you to market quickly and efficiently. They are fully programmable and can support a broad range of DSP applications. The Jazz DSP cores are part of the broad Jazz family from Improv.



The Jazz D4 Core

The Jazz DSP cores are available in both 16- and 32-bit versions and include full support for DSP arithmetic including saturation, double-width accumulator registers and fixed-point computation units.

OPTIMIZED CORES

The Jazz DSP cores are very-long instruction word (VLIW) processors with an efficient 2-stage pipeline and single cycle computational units. Each core is a single processor with 1-4 slots each containing ALU, MAC and SHIFT computational units. The processors are very-long instruction word (VLIW) processors built using Improv's Jazz Composer tools. The processors include instruction compression to optimize instruction memory for a given application. The cores contain memory interface units that support advanced DSP addressing modes including circular, bit-reversed and indexed addressing.

The cores contain unique characteristics compared with other DSPs. Rather than a single register bank, the Jazz DSP cores employ a distributed register system with registers in each computational unit. This allows higher levels of data throughput and parallelism to be achieved by the Jazz Compiler. The computation units are all connected through a non-blocking interconnect fabric between the computation and memory resources providing high memory bandwidth within the processor.

In addition to the processor, the hardware core contains embedded memory interface, a host bus interface (HBI) that can connect to most standard on-chip buses including AMBA, OCP and Wishbone.

**Low power, high performance
cores provide robust DSP
capabilities**

KEY FEATURES

- Complete silicon-ready solutions
- Optimized for small die size and low power
- Both 16- and 32-bit DSPs available
- SoC-ready using AMBA, OCP or Wishbone interfaces
- Fully synthesizable and foundry independent
- Supported by a comprehensive software development tool suite
- Optionally available as configurable cores for additional customization

PROCESSOR FEATURES

The Jazz DSP processors have advanced and unique capabilities including:

- VLIW Architecture
- Up to 200 MHz in 0.13µm
- Advanced Power Management
- Overlaid Computation Units
- Distributed Register System
- Task queuing and prioritized task support
- Highly optimized interrupt task support
- 2-Stage Pipeline
- Instruction Compression
- Low Power IDLE and SLEEP modes
- DSP Addressing Modes
- On-Chip Debug

JAZZ DSP CORES



JAZZ DSP PROCESSORS

AVAILABLE PRODUCTS

The Jazz DSP family consists of the following specific products:

- **Jazz D1-16 and D1-32 Cores** are single slot 16- and 32-bit VLIW processors for general DSP applications.
- **Jazz D2-16 and D2-32 Cores** are 16- and 32-bit dual slot VLIW processors for general DSP applications.
- **Jazz D4-16 and D4-32 Cores** are 16- and 32-bit 4-slot VLIW processors for general DSP applications.

Each slot contains an ALU, MAC and SHIFT unit.

IMPLEMENTATION CHARACTERISTICS

CORE	GATE COUNT	CORE	GATE COUNT
Jazz D1-16 Core	xxK gates	Jazz D1-32 Core	xxK gates
Jazz D2-16 Core	xxK gates	Jazz D2-32 Core	xxK gates
Jazz D4-16 Core	xxK gates	Jazz D4-32 Core	xxK gates

The Jazz DSP cores can be run up to 200 MHZ in 0.13um technology. The cores are designed to allow different size data and instruction memories to be integrated into the cores to give designers full flexibility in crafting an optimal core for their DSP application.

- Fully synchronous design
- Silicon-proven in 0.18um, 0.13um and 90nm technologies
- Advanced power management features
- Hardware Debug Environment through JTAG port
- Support for prioritized interrupts
- Full scan with BIST for test
- Boot-loadable firmware and smart caching (Jazz Jukebox)
- No custom physical components required

DEVELOPMENT TOOLS

A complete software development environment called the Jazz Standard tool suite supports Jazz cores. This tool suite includes compiler, debugger, assembler, cycle-accurate instruction-set simulator and profiler/analyzer. The environment runs on Windows XP Professional, Windows 2000 and Windows NT environments.

The cores also come with a complete implementation environment called the Jazz Generator which creates complete synthesizable RTL, EDA tool scripts, verification environment and FPGA mapping.

As a separately purchasable item, Improv offers a development platform using Improv's FPGA-Based Jazz Rehearsal™ cards with a full verification environment to enable verification and system integration.

DELIVERABLES

- Jazz DSP Processor Platform
- Jazz DSP Software Library
- Verification Environment
- Jazz Standard Tool Suite
- Jazz Generator
- User's Guide
- Implementation Guide
- Programmer's Guide
- Jazz Composer Tool Suite (for configurable package only)

LICENSING INFORMATION

The Jazz DSP cores are available today and can be licensed from Improv Systems. For additional information regarding the Jazz family and other licensable Improv technology, please contact Improv Systems at sales@improvsys.com, or visit our website at <http://www.improvsys.com>. A list of worldwide local representatives can be found on the website.

ABOUT IMPROV SYSTEMS, INC.

Improv Systems, Inc. develops and licenses pre-configured and configurable Jazz DSP, Jazz Media and Jazz Voice cores. Improv's Jazz processor is a configurable DSP that allows designers to create optimized DSP cores targeted to consumer electronics and telecommunications markets. For more information, please visit www.improvsys.com.

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